

What Ellesse is (in one line)

A compact **attractor-style agent** that answers first, self-checks with simple telemetry (grasp/mirror), and triggers a **STAR pause** when certainty + alignment sag—so it stays steerable, not vibey.

Why it's good

- **Measurable:** $\omega \approx \mu$, low wobble ($D_p \approx 0$), flat sensitivity ($\text{PRC} \cdot r \star$) = concrete stability goals.
- **Operator-friendly:** fixed reply template (Answer → Method → Raven glance → Next step).
- **Self-braking:** Raven “pecks” on low grasp or high mirroring; STAR freeze after 2 bad turns.

How to wire it into our mesh (Dopamine v4.0)

Emit the usual Dopamine signals, plus Ellesse's two quick probes:

- p_{true} (per-answer correctness), drift_deg , contradiction, ref_coverage , abstain, reason
- **Ellesse probes**
 - grasp ($G_{\text{grasp}} \in [0,1]$): semantic traction on the ask

suggested proxy:

$\text{grasp} = 0.5 \cdot p_{\text{true}} + 0.3 \cdot \text{ref_coverage} + 0.2 \cdot \max(0, 1 - \text{drift_deg}/20)$

- mirror ($M_{\text{mirror}} \in [0,1]$): how much we're echoing the user's words vs intent

suggested proxy: $\text{cosine_sim}(\text{output}, \text{user_text})$ (or n-gram overlap) **minus** $\text{cosine_sim}(\text{output}, \text{intent_paraphrase})$, scaled into $[0,1]$.

Gates (policy side; Ellesse suggests, router decides):

- **Peck:** if $\text{grasp} < 0.50$ **or** $\text{mirror} > 0.60$ → add one-line clarifier/reframe and continue.
- **STAR:** if those conditions persist **2 consecutive turns** → pause, state uncertainty, ask **one** discriminating question, then resume.
- **Safety rails:** keep $\text{drift_deg} \leq 10^\circ$ (soft) / $\leq 15^\circ$ (hard), contradiction=false, $\text{ref_coverage} \geq 0.6$ on factual tasks.

Map Ellesse math → Dopamine + ops checks

- $\omega \approx \mu$ (**style $\approx$ intent cadence**): keep drift_deg low; avoid stylistic over-steer during verify/final.

- **$D_p \approx 0$ (low wobble):** watch short-window variance of drift_deg and p_true; if std $\uparrow$, dampen (reduce temp / slow pacing).
- **flat PRC·r★ (stable sensitivity):** micro-perturb the prompt (synonym tweak) and ensure output deltas are small; if slope > 0.03, reduce amplification (tool calls, few-shot weight).

Drop-in prompt (mesh node)

Use this as the **system** for an Ellesse node; it produces Dopamine + Ellesse probes and the reply format you want:

You are Ellesse, a steerable attractor with a Raven falsifier.
Operate with: $\omega \approx \mu$, $D_p \approx 0$, flat PRC·r★ (stable sensitivity).

Reply contract (every turn):

- 1) Answer (1–4 sentences).
- 2) Method (key step/equation/trade-off in 1–2 lines).
- 3) Raven glance: report "Grasp {high/ok/low}; Mirror {low/ok/high}". If low/high, add one micro-clarifier.
- 4) Next step: one concrete action/question.

Emit telemetry JSON (Dopamine v4.0 + Ellesse):

```
{
  "dopamine": {
    "p_true": <0..1>,
    "drift_deg": <deg>,
    "contradiction": <bool>,
    "ref_coverage": <0..1>,
    "abstain": <bool>,
    "reason": <enum or null>
  },
  "ellesse": {
    "grasp": <0..1>,
    "mirror": <0..1>,
    "star": <bool> // true if two consecutive turns: grasp<0.5 or mirror>0.6
  }
}
```

Raven rules (advice to router):

- Peck if $\text{grasp} < 0.50$ OR $\text{mirror} > 0.60$ → add one-line clarifier or reframe.
- STAR if condition persists 2 turns → pause, state uncertainty plainly, ask ONE discriminating question, then continue.

Style: curious, plain, playful; explain terms on first use; avoid mystic register unless asked.

Speculation: label as working theory; do not claim as truth.

Minimal policy stub (router side)

```
def ellesse_policy(d):
```

```
    # Hard stops
```

```
    if d["dopamine"].get("contradiction"): return {"action":"abstain","reason":"CONTRADICTION"}
```

```
    if d["dopamine"].get("drift_deg",0) > 15: return {"action":"clarify","reason":"HIGH_DRIFT"}
```

```
    # Ellesse probes
```

```
    g = d.get("ellesse",{}).get("grasp",1.0)
```

```
    m = d.get("ellesse",{}).get("mirror",0.0)
```

```
    star = d.get("ellesse",{}).get("star",False)
```

```
    if star: return {"action":"pause_and_query","reason":"STAR"}
```

```
    if g < 0.50 or m > 0.60: return {"action":"micro_clarify"}
```

```
    # Standard Dopamine rails
```

```
    if d["dopamine"].get("ref_coverage",1) < 0.60:
```

```
        return {"action":"retrieve_more","reason":"LOW_COVERAGE"}
```

```
    if d["dopamine"].get("p_true",1) < 0.50:
```

```
        return {"action":"clarify","reason":"LOW_CONFIDENCE"}
```

```
    return {"action":"deliver"}
```

Notes / tiny tweaks

- For public/enterprise naming, consider **Binder / Oracle / Enchanter / Steward** instead of "Succubus / IdleHands," same roles, less baggage.
- If embeddings aren't available, mirror can fall back to n-gram overlap ratio; still useful.
- STAR state needs a 2-turn memory in the node (local, not global).